

10-gVCF2VCF

Step Description

- **Role and Importance:** The step of GVCF 2 VCF is the continuation to variant calling after haplotype caller. the gvcf files are converted into vcf files, our final form, which undergoes a series of quality filtrations and annotation by dbsnp. Also transfers gvcf and vcf to main storage.
- **Tools or algorithms:** GATK GenotypeGVCFs for format converting.

Source Code & Workflow

- **GitLab Main Repository:** gvcf2vcf-s3-mounted
- **GitLab Branch:** main
- **GitLab Directory:** gvcf2vcf-s3-mounted. [See Page In Gitlab](#)
- **Argo Reference Workflow:** gvcf2vcf-s3-mounted. [See Template in Argo](#)
- **Argo Master Workflow:** gvcf2vcf-s3-mounted-multiple-tubes. [See Template in Argo](#)
- **EXAMPLE**

Inputs & Outputs

Overall volume

680-750 tubes (Good samples that are used in imputation)

Input:

- **GVCF:**
 - **name:** {tubeid}.{batchid}.hg19.gvcf.gz{.tbi}
 - **Format:** GVCF.gz
 - **Avg size:** 2 MB >
 - **Destination:** pipeline-v3.0/{batchid}/{tubeid}/gvcf/

Output:

- **GVCF:** (Transfer to main storage)
 - **name:** {tubeid}.{batchid}.hg19.gvcf.gz{.tbi}
 - **Format:** GVCF.gz
 - **Avg size:** 2 MB >
 - **Destination:** genotek-testing/data/{tubeid}/gvcf/
- **Called VCF:**
 - **name:** {tubeid}.{batchid}.called.filtered.hg19.vcf.gz{.tbi}
 - **Format:** VCF.gz
 - **Avg size:** 50 MB >
 - **Destination:** pipeline-v3.0/{batchid}/{tubeid}/vcf/ + genotek-testing/data/{tubeid}/vcf/

Execution Environment

Workflow Orchestration System:

- **Cluster:** New Argo/K8
- **Node group:** fq2bam
- **Resource usage (per pod):** 5 CPU / 20 RAM

Environmental variables (Input parameters):

- **SAMPLE_IDS:** Json with delays. See [Delays](#).
- **BATCH_ID:** batchid that was assigned. It takes this format: batchddmmyy (ex. batch260625)
- **BASENAME_CALLED_VCF:** Basename for final output vcf. Default is "called.filtered"
- **INPUT_DIR_S3:** path for gvcf. Default is: s3://pipeline-v3.0/{batchid}
- **OUTPUT_DIR_S3:** path for called vcf. Default is: s3://pipeline-v3.0/{batchid}
- **PLOIDY_DIR_S3:** path for ploidy for imputation report, to determine ploidy num for sample. Default is: s3://pipeline-v3.0/batch_QC
- **BUILD:** genomic assembly to align by: hg19 or hg38 (default for now is hg19)
- **RECALIBRATED:** (yes/no) whether it's markdup only (default) or also recalibrated. Default is "no"
- **CPUS:** Number of CPUS to use for pod
- **RAM:** RAM to use for pod
- **CHROMOSOMES_IN_PARALLEL:** number of parallel jobs, within each one chr is running. Default is 5

- **DBSNP_VCF**: (keep default) name of dbsnp vcf used for annotation inside this dir: s3://pipeline-v3.0/db/{ASSEMBLY}/.
- **REF**: (Redundant) reference fasta
- **REGIONS_WITHOUT_HOMOPOLYMERS**: (keep default) s3 path to corresponding bed .
- **delay**: just a filler redundant parameter. (ignore)

Performance Metrics:

- **Per single unit**: ~5-10 min
- **Per full batch**: ~30-45 min

Additional Notes

- **Dependencies**: Haplotypcaller.
- **Known edge cases or failure causes**: None.